

Zhufeng (Zephyr) Qiu

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RESEARCH INTERESTS

GPU Systems | Compression-Communication Co-Design | HPC | ML Systems | Distributed LLM Training & Inference

RESEARCH SUMMARY

My background combines experience in production data pipelines and backend services with independent studies of multi-GPU execution, collective communication, KV-cache compression, and quantized LLM inference. Working with C++/CUDA, MPI/NCCL, and PyTorch, I use profiling and controlled experiments to find where simple performance models break down: the same optimization can win on one data path and lose on another. I aim to build distributed AI systems that stay efficient and correct across heterogeneous hardware.

EDUCATION

Northeastern University, Khoury College of Computer Sciences

Jan. 2023 - May 2025

- M.S. in Computer Science, GPA: 4.0/4.0
- Relevant coursework: Parallel Data Processing, Algorithms, Fundamentals of Computer Networking, Fundamentals of Cloud Computing, Database Management Systems.
- Teaching Assistant for Algorithms and Parallel Data Processing.

University of Southern California, Viterbi School of Engineering

Aug. 2019 - May 2021

- M.S. in Applied Data Science, GPA: 3.83/4.0
- Relevant coursework: Scientific Computing and Visualization, Analysis of Algorithms, Machine Learning for Data Science, Foundations and Applications of Data Mining, Database Systems.

Wuhan University, School of Resource and Environmental Sciences

Sep. 2014 - Jun. 2018

- B.S. in Geographical Information Science, GPA: 3.58/4.0
- Relevant coursework: Data Structures, Linear Algebra, Discrete Mathematics, Probability and Mathematical Statistics, C Programming.

Additional coursework: Operating Systems — Peking University via Coursera (non-credit) [Certificate](#)

Sep. 2026

SELECTED SYSTEMS, HPC & ML PROJECTS

Multi-GPU Similarity Engine - Lossless Homomorphic Collective Compression & Overlap [GitHub](#)

Jul. 2026 - Sep. 2026

Independent systems study extending USC INF 553 Data Mining / CSCI 596 Scientific Computing projects | C++, CUDA, NCCL, MPI,

OpenMP, PySpark, CMake, Nsight Systems

- Re-engineered a Yelp collaborative-filtering workload from Spark into **serial C++**, **OpenMP**, **MPI**, **CUDA**, and **dual-GPU NCCL** backends under a shared numerical contract; checked complete similarity outputs **within 1e-12** and verified identical retained pairs.
- Compressed the NCCL AllReduce **payload 3×** by packing six bounded integer sufficient statistics into two uint64 words with carry-free fields, allowing NCCL to **sum them directly in packed form** without decompression; verified exact integer-statistic reconstruction under explicit range constraints.
- Compared dimension and pair partitioning in FP64 on two A100 GPUs connected by NVLink with replicated, GPU-resident inputs. Pair partitioning **removed AllReduce** and reduced full resident-batch latency by **19.1% and 25.1%** on two views of the Yelp workload versus dimension partitioning; used **30 paired blocks** per view and included the complete output copy to CPU.
- Evaluated communication-computation overlap with **Nsight Systems** and chunk-size sweeps; derived the min(compute, comm)/total overlap ceiling, studied when overlap helps, and diagnosed and fixed a **double-buffering race**.

BoundRelay - LLM KV-Cache Compression and Transfer Study [GitHub](#)

Aug. 2026 - Sep. 2026

Independent ML-systems study | Python, PyTorch, CUDA, cuSZp, SZ3, CUDA zfp

- Built a PyTorch/CUDA benchmark around existing cuSZp, SZ3, and CUDA zfp compressors, covering **1,152 Qwen3-1.7B KV-tensor observations** with recorded model/codec revisions, frozen evaluation inputs, and a versioned corrections register.
- Ran **360 paired prepared-cache comparisons**: the selected configuration reduced payload to **0.330×** raw BF16 size and cut fsync-acknowledged write time by **26–49%**, but increased pipelined host-staged GPU transfer time to **1.52–2.33×** raw. Audited peer-copy failures that reported CUDA success without delivering data on three rented multi-GPU pods.
- Identified per-tensor overhead and payload-dependent bandwidth as sources of prediction error; stage costs measured on the actual cache and hardware predicted **21.43 ms versus 43.65 ms** for the complete pipeline.
- Replicated K/V compression-sensitivity asymmetry on **32 held-out articles**: at similar payload sizes, K-only compression raised mean teacher-forced NLL by **0.0155 (95% CI [+0.0068, +0.0235])**, while V-only compression did not; documented numerical-tolerance failures and a mid-round stop-rule deviation.

OPEN SOURCE CONTRIBUTION

SGLang-Omni - Fun-ASR Context-Budget Fix [PR #2296 \(open\)](#)

Sep. 2026

- Submitted a fix for Fun-ASR's language-dependent context-budget error to restore **30-second English audio admission**; added regression tests for default, Chinese, and English prompts and validated on an RTX 4090.

ADDITIONAL PROJECTS

Quantized LLM Inference Study on a LoRA-Fine-Tuned Llama 3.1 8B [GitHub](#) Jul. 2025 - Sep. 2025; Jul. 2026 - Aug. 2026

Independent inference-systems study extending an LLM-engineering course fine-tune | Python, PyTorch, CUDA, PEFT/LoRA, bitsandbytes, Hugging Face Transformers, Modal

- Measured a **2.50× generation-latency regression** at batch size one for the same NF4 Llama 3.1 8B model on T4 when using BF16 rather than FP16 compute. Separate phase measurements showed a **4.42× prefill slowdown**; unquantized FP16/BF16 generation on A10 differed by 0.8%.
- Measured NF4 decode running **1.69× slower** than unquantized FP16 on A10 despite a **3.9× smaller analytically estimated weight representation**; used memory-traffic and roofline estimates to examine why lower precision did not improve latency in this configuration.
- Built a **CUDA-event inference benchmark** across four inference configurations on T4 and A10, separating prefill and decode. For a 175-token prompt on T4 with NF4/FP16 compute, steady-state decode cost **93.5 ms/token versus 1.73 ms/token** averaged across prefill.
- Fine-tuned Meta Llama 3.1 8B with QLoRA (rank 32 / α 64 on attention projections, **27M trainable parameters, 0.34% of base**) under completion-only supervision on a curated 400,000-item dataset, and served it 4-bit NF4 on an NVIDIA T4 via Modal.

Zephyr Manus - Agent Runtime with Sandboxed Tool Execution [Live Demo](#)

Mar. 2026 - May 2026

Independent project | Python, FastAPI, Redis Streams, PostgreSQL, Docker, MCP/A2A

- Built a Planner-ReAct agent runtime with task decomposition, persistent state, and human-in-the-loop wait/resume; executed Browser, Shell, and File tools in Docker-isolated sandboxes, supported by an event-driven backend using Redis Streams, PostgreSQL, and SSE.

INDUSTRY EXPERIENCE

Beaconfire Solutions Inc. - Software Development Engineer

Jul. 2025 - Present

Seattle, WA (Remote)

- Developed enterprise HR and digital-commerce microservices, integrating authenticated APIs, relational/document data stores, AWS S3, and RabbitMQ-backed asynchronous workflows.

NSFOCUS Technologies Group Co., Ltd. - R&D Engineer

Jun. 2021 - Dec. 2022

Wuhan, China

- Built and maintained distributed security-data pipelines using Kafka and Elasticsearch for multi-source log ingestion and normalization, and implemented Spark/MapReduce batch workflows serving 100+ enterprise clients and processing approximately 5M records per day.
- Developed host-anomaly detection features for illicit file transfers and backdoor processes using fuzzy inference, hierarchical clustering, and k-core graph mining over multidimensional security logs.
- Contributed to a production alerting workflow that achieved effective response for **100% of severe threats and 90% of high-risk threats** in a telecom deployment.

NSFOCUS Technologies Group Co., Ltd. - R&D Engineer Intern

Jul. 2018 - Mar. 2019

Wuhan, China

- Processed and normalized multi-source security logs covering user operations, file access, and process behavior to support downstream host-anomaly analysis.

RESEARCH EXPERIENCE

PM2.5 Inversion and Spatiotemporal Analysis for the Wuhan Metropolitan Area

Mar. 2016 - Oct. 2017

National Undergraduate Innovation and Entrepreneurship Training Program | Core Team Member | Advisor: Prof. Huanfeng Shen

- Integrated ground-monitoring measurements, NASA MODIS/AOD satellite observations, and meteorological data, performing preprocessing and spatiotemporal matching across multiple data sources.
- Contributed to the development and evaluation of a mixed-effects model for PM2.5 inversion, using cross-validation and baseline comparisons to assess predictive performance.
- Used MATLAB/R for modeling and ArcGIS to generate spatially continuous PM2.5 concentration maps across the Wuhan “1+8” metropolitan area; the national program passed final review with a **Good** rating.

Web-Based Dynamic Thematic Mapping Technologies

Mar. 2018 - May 2018

Undergraduate Thesis | SinoMaps Press Collaboration | Advisors: Prof. Haihong Zhu and Yuntao Long

- Surveyed web-based thematic-mapping methods and designed the overall, functional, and logical architecture for online visualization, interaction, and dynamic data updates.
- Built and validated a D3.js customs trade-map and map-service prototype supporting thematic visualization, statistical charts, interactive queries, and live data updates.

TECHNICAL SKILLS

HPC / GPU: CUDA, NCCL, MPI, OpenMP, Nsight Systems, CMake

Languages: C/C++, Python, Scala, SQL

ML / LLM: PyTorch, Transformers, PEFT/LoRA, bitsandbytes, scikit-learn

Systems / Data: Spark, Hadoop, Kafka, Elasticsearch, Redis, SQL databases, AWS, Docker, Git

HONORS

- Outstanding Student Leader, Wuhan University, 2014-2015 and 2015-2016.
- Outstanding Individual in Theoretical Study, Wuhan University, 2014-2015.